(Revised curriculum) **Annex B**

MS Electrical Engineeirng - CEME, PNEC, SEECS, MCS & CAE

Course Code		Course Title	CHs
Core Courses			
EE	8XX	Linear System Theory	3
EE	891	Stochastic Systems	3
Solid State Electronics and Circuits			
EE	801	Semiconductor Device Physics	3
EE	802	Quantum Mechanics	3
EE	803	Physical Electronics	3
EE	804	Photonic Devices	3
EE	805	Semiconductor Processing	3
EE	806	Thin Film Processing	3
EE	807	Thin Film Characterization	3
EE	808	Digital Integrated Circuit Design	3
EE	809	Analog Integrated Circuit Design	3
EE	900	Optoelectronic Devices & Materials	3
EE	901	Power Electronics & Electric Drives	3
EE	902	Nano-Electronics	3
EE	903	Advanced Semiconductor Device Theory	3
EE	904	Microchip Fabrication Technology	3
EE	905	Advanced Power Electronics	3
EE	906	Solid State Electronics	3
EE	907	Micro & Nano Fabrication	3
EE	908	Ultra High Speed Nanoelectronic Devices	3
EE	909	Selected Topics in Electronics	3
EE	898	Nanotechnology	3
EE EC	823 825	Advanced Digital System Design	3
EE	826	Advanced VLSI Design	
EE	818	Micro-Electro-Mechanical Systems	3
EE	893	Data Acquisition & Mixed Signal Design	3
EE	847	Microwave Networks & Passive Components	
EE	942	Microwave Integrated Circuit Design	
NSE	821	Nano Fabrication by Self Assembly	3

NSE	845	Nano Lithography and Device Fabrication	3
MATH	816	Applied Linear Algebra	3
Electro - Medical Stream			
EE	811	Biomedical Imaging	3
EE	812	Medical Image Processing & Analysis	3
EE	813	Computational Pathology	3
EE	814	Cardiac Signal Processing	3
EE	815	Computational Molecular Biology	3
EE	816	Telemedicine Systems	3
EE	817	Tissue & Cell Engineering	3
EE	819	Biomedical Electro Mechanical Systems	3
EE	910	Biomaterials & Drug Delivery	3
EE	919	Selected Topics in Biomedical Engineering	3
CSE	888	Computational Modeling of Physiological Systems	3
MM	895	Bio-Informatics	3
BMES	811	Signals & Images in Medicine	3
BMES	812	Medical Devices Design & Standards	3
BMES	813	Biomedical Instrumentation	3
BMES	815	Biosensors and Instrumentation	3
BMES	821	Human Physiology and Anatomy	3
BMES	833	Prosthetics and Rehabilitation	3
BMES	842	Advanced Biomaterials	3
BME	822	Selected Topics in Biomedical Engineering	3
BME	831	Bio Fluid Mechanics	3
MATH	816	Applied Linear Algebra	3
Digital Systems			
EE EC	823 825	Advanced Digital System Design	3
EE MTS	821 841	Advanced Embedded System Design Advanced Embedded Systems	3 3
EE	826	Advanced VLSI Design	3
EE	822	ASIC Design Methodology	3
EE	808	Digital Integrated Circuit Design	3
EE	824	Real Time Systems	3
EE	825	System Level Packaging	3
EE	860	Mixed Signal IC Design	3
EE	828	Computerized Tomography Systems	3
EE	829	Digital Data Acquisition & Control	3
EE	920	System Validation	3

EE	921	System on Chip Architecture	3
EE	922	Design of Fault-Tolerant Systems	3
EE	929	Selected Topics in Digital Systems	3
EC	820	Advanced Computer Architecture	3
MATH	816	Applied Linear Algebra	3
Signal Processing			
EE CE	831 866	Advanced Digital Signal Processing	3
AV	820	Radar Signal Processing	3
EE	832	Pattern Recognition	3
EE	833	DSP Hardware System Design	3
EE	834	Applied Signal Processing	3
EE	835	Multirate Systems & Filter Banks	3
EE	836	Advanced Digital Image Processing	3
EE	837	Advanced Topics in Computer Vision & Image Processing	3
EE	838	Filtering & Tracking	3
EE	839	Adaptive Filters	3
EE	930	Spatial Array Processing	3
EE	931	DSP Software System Design	3
EE	932	Speech Processing	3
EE	933	Time Frequency Analysis	3
EE	939	Selected Topics in Signal Processing	
CS CE	867 803	Computer Vision	3
MATH	816	Applied Linear Algebra	3
RF & Microwave			
EE	847	Microwave Networks and Passive Components	3
EE	849	Electromagnetic Field Analysis	3
EE	840	RF MEMS: Theory and Applications	3
EE	841	Electromagnetic Theory	3
EE	842	Microwave Communication System Design	3
EE	843	Microwave Transmission Lines & Waveguides	3
EE	844	Antennas & Wave Propagation	3
EE	845	EMC/EMI	3
EE	846	Microwave Photonics	3
EE	848	Radiating Systems & Antennas	3
EE	940	Advanced RF Measurements	3
EE	941	RF Transceiver Design	3

EE	942	Microwave Integrated Circuit Design	3
EE	943	Microwave Devices I	3
EE	944	Microwave Devices II	3
EE	945	Computational Electromagnetics	3
EE	946	Advanced Antenna Theory and Design	3
EE	947	Microwave Devices & Systems	3
EE	948	Advance Electromagnetic Fields	3
EE	949	Selected Topics in Microwave Engineering	3
EE	896	Electrodynamics of Plasmas	3
EE	895	Analysis of Measurement Environment	3
MATH	816	Applied Linear Algebra	3
Smart Telecom Systems and Networks			
EE	851	Advanced Digital Communication Systems	3
EE	852	Information & Coding Theory	3
EE	853	Advanced Wireless Communication	3
EE	854	Optical Communication Systems	3
EE	855	Error Control Coding	3
EE	856	Software Defined Radio	3
EE	857	Advanced Satellite Communication Systems	3
EE	858	Communication Project Management	3
EE	859	Performance Analysis of Communication Networks	3
EE	950	Advanced Data Communication Systems	3
EE	951	Radar Systems	3
EE	959	Selected Topics in Communication Systems	3
EE	897	Detection & Estimation	3
EE CE	831 866	Advanced Digital Signal Processing	3
CSE	812	RF Communication System Design	3
EE	881	Advanced Communication Networks	3
EE	882	Cognitive Radio Networks	3
EE	883	Wireless Sensor & Mesh Networks	3
EE	884	Photonic Networks	3
EE	885	Broadband Networks	3
EE	886	Advanced Wireless Networks	3
EE	887	Network Switching & Routing	3
EE	888	Advanced Computer Network Design & System Security	3
EE	889	Network and Service Management and Control	3
EE	989	Selected Topics in Networks	3
CSE	820	Advanced Computer Networks	3
CS	879	Network Performance Analysis	3

MATH	816	Applied Linear Algebra	3
Power			
EE	863	Power System Analysis	3
EE EEE	862 800	Power System Operation & Control OR Power System Operation, Control and Optimization	3
EEE	801	Clean Energy Generation, Integration and Storage	3
EEE	802	Advanced Power System Stability and Transient Studies	3
EEE	811	Electric Power Quality	3
EEE	812	Advanced Power System Protection	3
EEE	813	Computer Modelling of Electrical Power Systems	3
EEE	814	Advanced Power Electronics	3
EEE	815	Electric Power Generation Transmission and Distribution	3
EEE	816	Electric Power Markets	3
EEE	817	High Voltage Engineering	3
EE	861	Alternating Current Electrical Machines and Drives	3
ESE	803	Photovoltaic Devices	3
ESE	813	Energy Economics and Policy	3
ESE	814	Fuel Cells	3
ESE	817	Wind Energy	3
ESE	820	Energy and Environment	3
ESE	824	Nuclear Energy Engineering	3
ESE	835	Materials Science and Engineering	3
ESE	836	Li-ion and Next Generation Batteries	3
ESE	837	Electrochemical Energy Storage and Conversion	3
ESE	838	Development and Evaluation of Energy Projects	3
ESE	909	Smart Grid Architecture	3
EE	861	Alternating Current Electrical Machines and Drives	3
EE	864	Advanced Machines	3
EE	865	Power Generation Operation & Control	3
EPE	903	Application of Optimization Methods in Electrical Power Systems	3
EEE	904	Special Topics in Dielectrics and Electrical Insulators	3
EEE	900	Signal Processing for Modern Power Systems	3
EEE	901	Advanced Converter Control Technique	3
EEE	902	Technologies for Smart Transmission Systems	3
ESE	910	Smart Power Systems	3
ESE	811	Solar Energy	3
IS	838	Advanced Simulation and Modelling	3
CS	877	Artificial Intelligence and Machine Learning	3
MATH	901	Advanced Engineering Mathematics	3

ESE	831	Energy Policy Analysis and Planning	3
ESE	832	Energy and Climate Change	3
ESE	833	Industrial Energy Management	3
ESE	834	Sustainable Buildings	3
EE	969	Selected Topics in Power Systems	3
MATH	816	Applied Linear Algebra	3
Control Systems			
EE	871	Linear Control Systems	3
AV	814	Flight Dynamics and Controls	3
EE	872	Optimal Control	3
EE	873	Fuzzy Control	3
EE	874	Adaptive Control	3
EE	875	Discrete Time Control Systems	3
EE	876	Probabilistic Robotics	3
EE	877	Mobile Robotics	3
EE	878	System Identification	3
EE	879	Robust Control	3
EE	970	Advanced Robotics	3
EE ME	971 816	Modeling & Simulation of Dynamic Systems Modeling and Simulation of Dynamic Systems	3 3
EE	972	Advanced Digital Control Systems	3
EE	973	Control System Optimization	3
EE	974	Networked & Embedded Control Systems	3
EE	975	Robust & Multivariable Control Systems	3
EE	976	Optimal & Multivariable Control Systems	3
EE	977	Nonlinear Control Systems	3
EE	978	Convex Optimization	3
EE	979	Selected Topics in Control Systems	3
EE	894	Cognitive Robotics	3
EE	892	Instrumentation & Systems	3
EM	800	Robotics - 1	3
EM	805	Robotics - 2	3
MTS	800	Advanced Robotics I	3
MTS	801	Advanced Robotics II	3
MTS	840	Data Acquisition and Control	3
ME	837	Nonlinear Dynamics	3
ME	812	Advanced Control Systems-I	3
MATH	816	Applied Linear Algebra	3
AI & Autonomous Systems			

EE	836	Advanced Digital Image Processing	3
EE	876	Probabilistic Robotics	3
EE	877	Mobile Robotics	3
EE	837	Advanced Topics in Computer Vision & Image Processing	3
EE	897	Detection & Estimation	3
EE	970	Advanced Robotics	3
EE	839	Adaptive Filters	3
EE	832	Pattern Recognition	3
EE	871	Linear Control Systems	3
EE	821	Advanced Embedded System Design	3
EE	878	System Identification	3
EE	985	Graphical Models for Autonomous System	3
SE	801	Artificial Neural Networks	3
SE	807	Machine Learning	3
CS	867	Computer Vision	3
SYSE	804	Modeling, Simulation & Optimization	3
MATH	816	Applied Linear Algebra	3
Energy Policy & Management			
Energy Policy & Economics Pool			
CE	886	Water Resources, Economics, Planning and Management	3
ECO	682	Economics of Environment and Natural Resources	3
ECO	704	Applied Microeconomics	3
ECO	726	Applied Econometrics	3
ECO	932	Development Policy and Planning	3
EEE	816	Electric Power Markets	3
ESE	813	Energy Economics and Policy	3
ESE	831	Energy Policy Analysis and Planning	3
ESE	838	Development and Evaluation of Energy Projects	3
Energy Management Pool			
ESE	819	Environmental Impact Assessment	3
ESE	820	Energy and Environment	3
ESE	832	Energy and Climate Change	3
ESE	833	Industrial Energy Management	3
ESE	834	Sustainable Buildings	3
Integrated Circuits and Systems			
EE	808	Digital Integrated Circuit Design	3
EE	809	Analog Integrated Circuit Design	3
EE	903	Advanced Semiconductor Device Theory	3

EE	904	Microchip Fabrication Technology	3
EE	822	ASIC Design Methodology	3
EE EC	823 825	Advanced Digital System Design	3
EE	826	Advanced VLSI Design	
EE	860	Mixed Signal IC Design	3
EE	921	System on Chip Architecture	3
EE	821	Advanced Embedded System Design	3
EC	820	Advanced Computer Architecture	3
CS	853	Formal Methods	3
EE	820	System Validation	3
EE	941	RF Transceiver Design	3
EE	940	Advanced RF Measurements	3
EE	840	RF MEMS: Theory and Applications	3
EE	818	Micro-Electro-Mechanical Systems	3
CSE	812	RF Communication System Design	3
EE	847	Microwave Networks & Passive Components	3
EE	942	Microwave Integrated Circuit Design	3
EE	825	System Level Packaging	3
EE	947	Microwave Devices & Systems	3
EE	866	Advanced Analog IC Design	3
EE	867	Radio Frequency Integrated Circuit Design	3
EE	868	mm-Wave and Terahertz Integrated Circuit Design	3
EE	880	Photonic Integrated Circuits	3
EE	960	Integrated Circuits Packaging	3
EE	961	Signal Integrity in High-Speed Designs	3
EE	962	Integrated Circuit Design-for-Test	3
EE	963	Hardware Security	3
EE	964	Special Topics in Integrated Circuits and Systems	3
		Thesis / Research	
EE	899	MS Thesis	6
RM	898	Research Methodology	2
SEM/WKSP	897	Seminar / Workshop	1
Courses of other Departments(Information / Systems / Computational & Software Engineering) approved for MS/PhD Electrical Engineering			
MATH	812	Advanced Engineering Mathematics	3
MATH	816	Applied Linear Algebra	3
MATH	850	Advanced Numerical Analysis	3
IS	820	Computer Security	3

IS	822	Wireless Network Security	3
IS	827	Electronic Warfare – Principles and Techniques	3
IS	851	Secure Communications	3
IS	852	Data Communication Networks & Security	3
IS	855	Information Hiding	3
IS	863	Cellular and Mobile Network Security	3
IS	893	Micro & Nano Fabrication	3
SYSE	804	Modeling, Simulation & Optimization	3
SYSE	844	Communication System Engineering	3
SYSE	845	Advanced Communication System and Network Security	3
SYSE	861	Introduction to C4I2SR System Engineering	3
SYSE	862	Architecting C4I System of Systems	3
CS	882	Advanced Information Security	3
CSE	812	Distributed Systems and Resource Optimization	3
CSE	842	Communication Systems & Networks	3
CSE	843	Performance Analysis of Communication Systems	3
CSE	844	Performance Analysis of Networks	3
SE	807	Machine Learning	3
SE	801	Artificial Neural Networks	3
SE	807	Machine Learning	3
SE	808	Bio Informatics Systems	3
SE	826	Advanced Computer Network Design and System Security	3
SE	828	Network Security	3
SE	851	Wavelet Compression	3
SE	865	Human computer Interface	3
SE	898	Research Methodologies	3